

Foundation (Jalapeno)

- Q1.** Which of the following is the graph of the inequality $y > 2x - 3$?
- (A) Solid line with shading above
(B) Dashed line with shading below
(C) Solid line with shading below
(D) Dashed line with shading above
(E) No shading
- Q2.** The inequality $x + 2y < 5$ has a boundary line that is
- (A) Solid
(B) Dashed
(C) Horizontal
(D) Vertical
(E) Diagonal
- Q3.** A company's profit is given by the inequality $P > 200x - 1000$, where x is the number of products sold. Which region of the graph represents the company's profit?
- (A) Above the line
(B) Below the line
(C) On the line
(D) To the left of the line
(E) To the right of the line
- Q4.** The inequality $y \leq -x + 2$ represents a region that is
- (A) Bounded by a solid line
(B) Bounded by a dashed line
(C) Unbounded
(D) Bounded by a horizontal line
(E) Bounded by a vertical line
- Q5.** A person's savings account balance is given by the inequality $B < 1000 - 50x$, where x is the number of months. Which of the following is true?
- (A) The balance increases over time
(B) The balance decreases over time
(C) The balance remains constant
(D) The balance is always positive
(E) The balance is always negative
- Q6.** The inequality $2x + 3y > 7$ represents a region that is
- (A) Bounded by a solid line
(B) Bounded by a dashed line
(C) Unbounded
(D) Bounded by a horizontal line
(E) Bounded by a vertical line
- Q7.** A stock's value is given by the inequality $V > 50x - 200$, where x is the number of days. Which region of the graph represents the stock's value?
- (A) Above the line
(B) Below the line
(C) On the line
(D) To the left of the line
(E) To the right of the line
- Q8.** The inequality $y \geq x - 2$ represents a region that is
- (A) Bounded by a solid line
(B) Bounded by a dashed line
(C) Unbounded
(D) Bounded by a horizontal line
(E) Bounded by a vertical line

Open Ended Questions (FRQ)

- Q9.** Graph the inequality $x + 2y < 4$ and shade the region that satisfies the inequality. Find the x - and y -intercepts.
- Q10.** A company's tax liability is given by the inequality $T > 0.2x - 1000$, where x is the company's income. Graph the inequality and shade the region that satisfies the inequality. Find the income level at which the tax liability is 0.

Chef's Tips

- Tip 1: Emphasize the importance of shading the correct region
- Tip 2: Encourage students to check their work by plugging in test points
- Tip 3: Remind students to label their axes and include a title on their graph